Cloud Networking

private by default, public by exception

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In one line: put workloads in a private **virtual network** split into tiered subnets, expose only the load balancer, allow each tier only from the tier above with **security groups**, reach managed services through **private endpoints**, send outbound traffic through a **NAT gateway**, and watch the data-transfer bill.

1 The database with a public IP

A team launches its first cloud app. The database gets a public IP so developers can connect from home, every VM can talk to every other VM, and outbound traffic leaves however the cloud defaults allow. It works until a security scan finds Postgres listening on the internet, protected only by a password. Internet-wide scanners find open database ports within hours. Good cloud networking makes that state impossible by construction.

2 Virtual networks, address space and subnets

A **VPC** (AWS, Google Cloud) or **VNet** (Azure) is your private address space, normally from the RFC 1918 ranges: 10.0.0.0/8, 172.16.0.0/12, 192.168.0.0/16. A `/16` such as 10.0.0.0/16 holds 65,536 addresses; a `/24` subnet holds 256, of which AWS and Azure each reserve 5 (the first four and the last: network address, gateway, DNS and other provider uses, and broadcast), so 251 are usable. Split by tier:

- **Public subnet:** only load balancers, application gateways and NAT. On AWS a subnet is "public" because its route table sends 0.0.0.0/0 to an internet gateway.
- **Private subnet:** app servers, containers, Kubernetes nodes. No public IPs.
- **Data (isolated) subnet:** databases and private endpoints, reachable only from the app subnet.

A zonal detail: an AWS subnet lives in one availability zone, so you create one subnet per tier **per AZ**; an Azure subnet spans the region's zones. Size generously: Kubernetes pods that take VNet IPs (Azure CNI, AWS VPC CNI) can consume thousands of addresses (lesson 60).

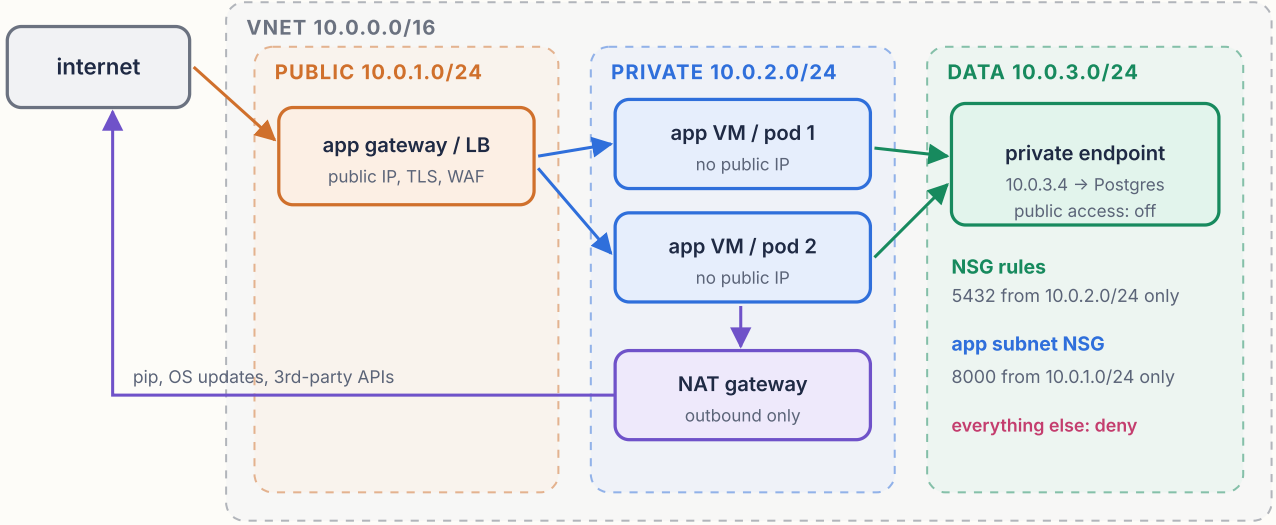


Figure 1. A three-tier layout. Only the gateway has a public IP; each NSG allows traffic only from the tier above; outbound goes through NAT.

3 Security groups: allow lists

Security groups (AWS) and **network security groups** (Azure) are stateful firewalls attached to resources or subnets: allow the inbound connection and the reply is allowed automatically. Write them as allow lists by tier: the app tier accepts port 8000 only from the load balancer; the database accepts 5432 only from the app tier. AWS lets a rule reference another security group ("from sg-app") instead of IP ranges, and Azure has application security groups for the same purpose, so rules survive scaling and IP changes.

Know the defaults. An Azure NSG has default rules that **allow all traffic inside the VNet** (AllowVNetInBound, priority 65000) before the final deny, so tier isolation needs explicit deny or narrow allow rules at lower priority numbers. AWS also has stateless **network ACLs** at subnet level, a coarse second layer where you must allow return traffic yourself.

4 NAT gateways and private endpoints

Private resources still need outbound access: package updates, container image pulls, third-party APIs. A **NAT gateway** lets them open connections out while remaining unreachable from outside, and gives all outbound traffic a fixed public IP that partners can allow-list. On AWS a NAT gateway is zonal, so production uses one per AZ.

Managed services (databases, storage, key vaults, queues) have public endpoints by default. A **private endpoint** (Azure Private Link, AWS PrivateLink interface endpoints) gives the service a private IP inside your subnet, so traffic never crosses the internet, and you can then **disable public access** on the service entirely. The catch is DNS: the service's normal hostname must resolve to the private IP inside your network, via a private DNS zone such as `privatelink.postgres.database.azure.com`, or clients silently keep using the public endpoint. On AWS, **gateway endpoints** for S3 and DynamoDB are free and keep that traffic off the NAT gateway.

5 Connecting networks: peering and hub-and-spoke

Peering links two virtual networks directly with private routing. It is **not transitive**: if A peers with B and B with C, A cannot reach C through B. With many networks, use a **hub-and-spoke** design: a central hub holds shared services (firewall, VPN or ExpressRoute / Direct Connect gateway to on-premises, DNS), and each workload spoke peers only with the hub. On AWS a **Transit Gateway** plays the hub role as a managed router; on Azure, a hub VNet or Virtual WAN.

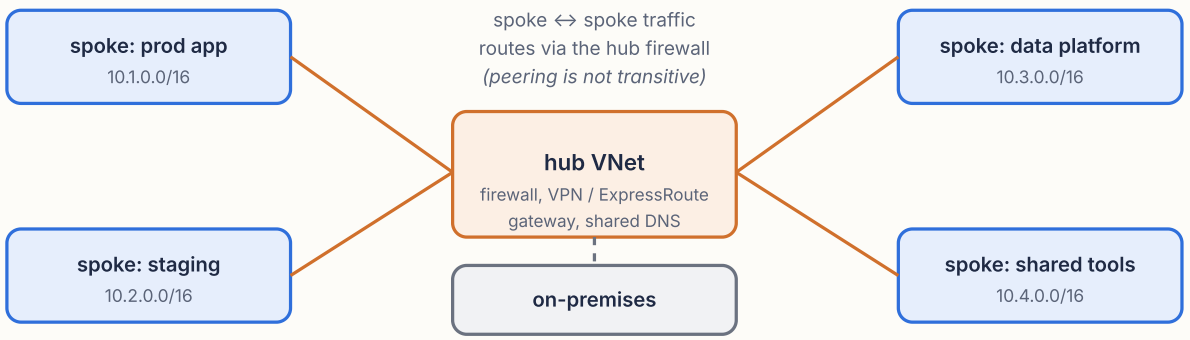


Figure 2. Hub-and-spoke: one place for shared firewalls, inspection and on-premises connectivity.

Plan address ranges early. Overlapping CIDR ranges cannot be peered or routed, and renumbering a live network is painful. Keep an IP address plan (e.g. a /16 per spoke from a reserved 10.0.0.0/8), and never reuse the default range in every environment.

6 What network traffic costs

Traffic	Typical AWS list price	Notes
Within one AZ (private IPs)	free	keep chatty services together
Between AZs in a region	$0.01/GB each direction	replication and cross-zone calls add up
Between regions	about $0.02/GB (varies by pair)	cross-region replication, DR
Out to the internet	about $0.09/GB for the first 10 TB/month	first 100 GB/month free; a CDN is usually cheaper
NAT gateway	$0.045/hour + $0.045/GB processed (us-east-1)	on top of transfer charges

Prices vary by provider and region (Azure and Google Cloud have their own tables), but the shape is the same: inside a zone is cheap, crossing zones, regions and the internet costs more each step. Classic surprises: a service pulling large container images or S3 data through the NAT gateway, chatty microservices spread across zones, and large downloads served from origin instead of a CDN. Cost optimisation is lesson 64.

WORKED EXAMPLE: AN APP REACHING A MANAGED DATABASE PRIVATELY

VNet `10.0.0.0/16` has three subnets: `10.0.1.0/24` for Azure Application Gateway (the only public IP), `10.0.2.0/24` for the app servers, and `10.0.3.0/24` holding a private endpoint for Azure Database for PostgreSQL at `10.0.3.4`. Public network access on the database is **disabled**. A private DNS zone makes the server's hostname resolve to 10.0.3.4 inside the VNet. The data subnet's NSG allows 5432 only from 10.0.2.0/24 and denies other VNet traffic; the app subnet allows 8000 only from the gateway subnet. A NAT gateway on the app subnet handles outbound calls. The security scan now finds one open port on the internet: 443 on the gateway.

COMMON MISTAKES

- Public IPs on databases or VMs "just for debugging" (use a bastion or VPN).
- Relying on Azure's default NSG rules, which allow all traffic inside the VNet.
- Private endpoint created but DNS still resolving to the public address.
- Overlapping address ranges that block peering later.

7 Key terms

VPC / VNet: private address space in the cloud

Security group / NSG: stateful allow-list firewall

Private endpoint: private IP for a managed service

Subnet: a slice of the range, usually one per tier

NAT gateway: outbound-only internet access for private resources

Hub-and-spoke: shared hub network that spokes peer with

8 Interview questions

Q1. Design a secure network layout for a three-tier app in the cloud. BASIC
One virtual network with public, private and data subnets; only the load balancer is public; security groups allow each tier only from the tier above; managed services via private endpoints; a NAT gateway for outbound access.

Q2. What drives network cost? BASIC
Cross-zone, cross-region and internet egress traffic, plus NAT gateway processing; traffic within a zone is usually free.

Q3. How many usable addresses does a /24 subnet have on AWS or Azure? BASIC
251: 256 minus the 5 addresses each provider reserves.

Q4. A private app server needs to call a third-party API that allow-lists IPs. How? SCENARIO
Route outbound traffic through a NAT gateway with a static public IP and give that IP to the partner.

Q5. You added a private endpoint for your database, but traffic still goes over the public endpoint. Why? SCENARIO
DNS: the hostname still resolves to the public IP; link a private DNS zone (privatelink...) to the VNet so it resolves to the private endpoint.

Q6. VNet A peers with B, and B with C. Can A reach C? TRICKY
No, peering isn't transitive; use a hub with a firewall or router (or Transit Gateway / Virtual WAN), or peer A and C directly.

Q7. Your NAT gateway bill is larger than your compute bill. What's likely happening? TRICKY
Heavy traffic such as S3 reads or image pulls is going through NAT; use gateway or private endpoints for cloud services and cache or mirror images.

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